Model card: three DeBERTa-v3-xsmall models for RTE

[Student name] · [Roll number] · CS6315E RTE Challenge
code: submitted alongside (see the README)

Architecture (shared by all three). `microsoft/deberta-v3-xsmall` encoder (12 layers, hidden 384, 6 heads) + pooler + linear 2-way head; full fine-tuning, **no** LoRA/adapters. Model B additionally has a *vocabulary-pruned* input embedding: only the rows of 62,245 tokens are kept (special tokens ∪ every training-set token ∪ the 60,000 most frequent tokens of the tokenizer's unigram model); other token ids map to `[UNK]` through a fixed lookup table (int32, 0.5 MB, not a parameter).

The three models.

	Training	Params	Val	Dev
A	masked KD from DeBERTa-v3-base, seed 21 (epoch 7)	70,830,722	82.7	78.0
B	model A, vocabulary pruned after training	45,542,402	79.8	78.3
C	plain fine-tuning on gold labels, seed 21	70,830,722	81.0	80.9

All three: 21,640,322 non-embedding parameters; word embedding $128{,}100 \times 384 = 49{,}190{,}400$ (A, C) or $62{,}245 \times 384 = 23{,}902{,}080$ (B). All parameters trainable. Val: premise-grouped 10% split of train (B: with only the fit split's tokens, so the text is unseen).

Why these three. A has the highest validation accuracy of all 20 xsmall checkpoints (4 methods × 5 seeds). B removes the 36% of parameters that sit in embedding rows RTE rarely uses, at an estimated 0–3 points. C is the best checkpoint trained without the teacher, so its errors differ from A's. Accuracy first: A; accuracy per parameter: B.

Training data. RTE train (2,490 pairs; identical to GLUE RTE train), minus the validation split. Teacher soft labels (A, B) are out-of-fold logits of five DeBERTa-v3-base models trained on the same data. No external NLI/RTE data, no LLM-generated data, no pre-fine-tuned RTE checkpoints. Dev (GLUE RTE validation, 277) is used only for reporting.

Training hyperparameters.

Loss	A, B: $0.5\,\mathrm{CE} + 0.5\,T^2\,\mathrm{KL}$, T=2, pure CE where the teacher is wrong; C: CE
Optimiser	AdamW, weight decay 0.01, grad-clip 1.0
Learning rate	5×10^{-5} top layer, layer-wise decay 0.9; head 1×10^{-4}
Schedule / batch / length	10% linear warm-up then decay / 16 / 256 tokens
Epochs	up to 15, early stopping on validation accuracy, patience 4
Precision / hardware	bf16 autocast on Colab T4 (reruns bit-identical); PyTorch 2.14, Transformers 5.22.2

Reference. DeBERTa-v3-base (teacher), 184.4M parameters: dev 84.8 ± 1.1 over 5 seeds.

Limits. Short English news-style pairs. Weak on high-overlap non-entailments, numbers and modality. B loses tokens outside its vocabulary to `[UNK]` (98.5% of dev tokens kept).

Run. `python scripts/predict_test.py -input <test file>` runs all three (README).